

HashMap Problem Bank

Legend

★ = Core Placement Problem (Mandatory)

Language-Agnostic DSA Problem Bank | Focus: HashMap as an algorithmic tool. Understand hashing internals.

Section 1: HashMap Internals — Implementation

★ **Design a Simple Hash Function for Integer Keys**

★ **Implement HashMap using Array of Linked Lists (Chaining)**

Implement HashMap using Open Addressing (Linear Probing)

★ **Insert, Search, and Delete in a Custom HashMap**

Handle Collision using Quadratic Probing

★ **Implement Rehashing when Load Factor Exceeds Threshold**

Display All Key-Value Pairs in a Custom HashMap

Section 2: Frequency and Counting Problems

★ **Count Frequency of Each Element in an Array**

★ **Find the Most Frequent Element in an Array**

★ **Find the Least Frequent Element in an Array**

★ **Count Frequency of Each Character in a String**

Find the First Non-Repeating Character

★ **Find the First Repeating Element in an Array**

Find All Elements Appearing More than $N/3$ Times

Section 3: Pair and Sum Problems

★ **Two Sum — Find Pair with Given Sum**

★ **Two Sum — Return Indices of the Pair**

Three Sum — Find All Triplets with Zero Sum

★ **Four Sum — Find All Quadruplets with Given Target**

Count Pairs with Given Difference K

★ **Count Pairs with Given Sum**

Find All Pairs with Equal Product

Section 4: Subarray Problems using HashMap

★ **Subarray with Zero Sum (Exists or Not)**

★ **Count Subarrays with Zero Sum**

★ **Longest Subarray with Equal Number of 0s and 1s**

★ **Subarray with Given Sum using HashMap (Prefix Sum)**

★ **Count Subarrays with Given Sum**

Longest Subarray with Sum Divisible by K

Section 5: String and Anagram Problems

- ★ **Check if Two Strings are Anagrams**

- ★ **Group Anagrams Together from a List of Strings**

Find All Anagram Substrings in a String

- ★ **Longest Substring without Repeating Characters**

Minimum Window Substring Containing All Characters

Count Distinct Substrings using Hashing

Section 6: Advanced HashMap Patterns

- ★ **Isomorphic Strings Check using HashMap**

- ★ **Word Pattern Match (e.g. 'abba' maps to words)**

- ★ **Find Longest Consecutive Sequence in an Array**

- ★ **Intersection and Union of Two Arrays using HashMap**

Clone a Graph using HashMap

- ★ **Implement LRU Cache using HashMap and Doubly Linked List**

Check if Array can be Divided into Pairs with Equal Sum

Section 7: Real-World Applications of HashMap

- ★ **Student Grade Lookup System — Store and Retrieve Grades by Student ID**

- ★ **Phone Directory — Map Contact Names to Phone Numbers**

- ★ **Word Frequency Counter in a Document — Find Top K Frequent Words**

Inventory Stock Tracker — Map Product ID to Quantity

- ★ **URL Shortener — Map Short Code to Original URL using Hashing**

Duplicate File Finder — Group Files by Hash of Content

- ★ **Login Session Manager — Map Session Token to User ID**

Spell Checker — Store Dictionary Words in HashMap for O(1) Lookup